Session 4: More on Regression and Graph

Various Regression, table export, Graphs plotting

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Introduction

What this session is about

Explore different regression models, and different types of graphs, and present the result clearly.

- Review causal reasoning through six short slides.
- Extend `lm()` using interactions and curves; introduce `feols()` and IV.
- Compare and export results using several table tools.
- Extend last session's graphs and save publication-ready outputs.

Lecture Review

Lecture Review — Association is not a policy effect

Policy question: Would building a park improve mental health?

- Treatment: access to a park. Outcome: mental-health score.
- Income can influence both park access and mental health **(confounding variable)**.
- Existing mental health may influence residential choice **(reverse causality)**.
- These processes can coexist with a real park effect.

Lecture Review — potential outcomes and the missing

For the same person, define:

- $Y_i(1)$: mental health with a park.
- $Y_i(0)$: mental health without a park.
- Individual causal effect: $Y_i(1) - Y_i(0)$.

We observe one potential outcome per person. The other is the counterfactual.

The **Average Treatment Effect on the Treated** averages this causal effect among those who actually receive treatment.

Lecture Review — observed data gap = ATT + selection

$$\underbrace{E[Y \mid D=1] - E[Y \mid D=0]}_{\text{Observed gap}} = \underbrace{E[Y(1) - Y(0) \mid D=1]}_{\text{ATT}} + \underbrace{E[Y(0) \mid D=1] - E[Y(0) \mid D=0]}_{\text{Selection bias}}$$

For example: 10-point observed gap = 4-point park effect + 6-point baseline difference.

- ATT compares the **same group** under two treatment conditions.
- Selection bias compares **different groups** under no treatment.

Lecture Review — controls and randomization

- Controls can help reduce confounding.
- Unobserved factors matter when they influence assignment and potential outcomes.
- Random allocation makes treatment independent of baseline characteristics in expectation.
- Statistical significance, extra controls, and higher R^2 do not establish causality.

More on Regression

Review: select needed data and run basic regression

```
WHR2016 ← whr_panel %>% filter(year == 2016) %>%
  select(country, region, happiness_score, economy_gdp_per_capita,
         health_life_expectancy, freedom) %>% drop_na()

census ←readRDS (here("Datawork","Datasets","Final","census.RDS"))


m1 ← lm(happiness_score ~ economy_gdp_per_capita, data=WHR2016)
m2 ← lm(happiness_score ~ economy_gdp_per_capita +
           health_life_expectancy + freedom, data=WHR2016)

models ← list("gdp/capita only"=m1, "with controls"=m2)
```

Regression with lm()

Formula	Meaning
`y ~ x1 + x2`	Additive associations
`y ~ .`	All other columns in the supplied data
`y ~ x1 * x2`	`x1 + x2 + x1:x2`
`y ~ x1:x2`	Interaction term only, plus the default intercept
`y ~ x + I(x^2)` or `y ~ poly(x,2,raw = TRUE)`	Quadratic relationship
`y ~ x + factor(region)`	Region-specific intercepts

With all selected predictors: lm(y ~ .)

Usually, we select columns first. And then use `.` which would include all those country identifiers, year, or outcome-related variables you intend to test on and control for.

```
reg_data ← WHR2016 %>%
  select(happiness_score, economy_gdp_per_capita,
         health_life_expectancy, freedom)
m_all ← lm(happiness_score ~ ., data=reg_data)
coef(m_all)
```

Interactions: lm(y ~ x1 * x2)

The interaction is the association between one independent variable and the output depends on another variable. (relationship between x_1 and y changes with x_2) For example, the relationship between income and happiness might differ by freedom level.

Covariance between x_1 and x_2 won't tell you whether or not should include an interaction. (In another word, uncorrelated variables can interact.)

`economy_gdp_per_capita * freedom` includes both main effects and their interaction.

The GDP slope here is $\beta_{gdpperpcaita} + \beta_{interaction}\, freedom$.

```
m_interact ← lm(happiness_score ~ economy_gdp_per_capita * freedom +
              health_life_expectancy, data=WHR2016)
coef(m_interact)
```

Interactions: lm(y ~ x1 * x2)

An interaction with a dummy variable allows a relationship to differ between two groups. For example, if you want to know how education relates to earnings differ between male and female group. It is the time to include in a dummy variable Female (female=1, male=0). The estimation would look like:

$$Earnings = \beta_0 + \beta_1 Education + \beta_2 Female + \beta_3 (Education \times Female) + \varepsilon$$

Substituting the dummy variable's two values gives:

Group	Predicted earnings
Men (`female = 0`)	$\beta_0 + \beta_1 Education$
Women (`female = 1`)	$(\beta_0 + \beta_2) + (\beta_1 + \beta_3) Education$

Therefore:

- β_2: the difference between women and men at **zero years of education**.
- β_3: the difference between their **education slopes**.

We will see later that Difference-in-Difference method, a particular application of interactions, shares the similar interaction structure.

Interactions: lm(y ~ x1:x2)

`:` alone omits the two numeric main effects. Usually retain main effects unless the restriction has a substantive justification.

```
m_product ← lm(happiness_score ~
                 economy_gdp_per_capita:freedom,
               data=WHR2016)
m_interact ← lm(happiness_score ~
                 economy_gdp_per_capita * freedom,
               data=WHR2016)
modelsummary(list("Product only"=m_product,
                  "Main effects + product"=m_interact),
             output="kableExtra")
```

	Product only	**Main effects + product**
(Intercept)	3.982	3.614
	(0.089)	(0.360)
economy_gdp_per_capita × freedom	3.619	2.400
	(0.191)	(0.880)
economy_gdp_per_capita		0.830
		(0.406)
freedom		0.315
		(0.861)

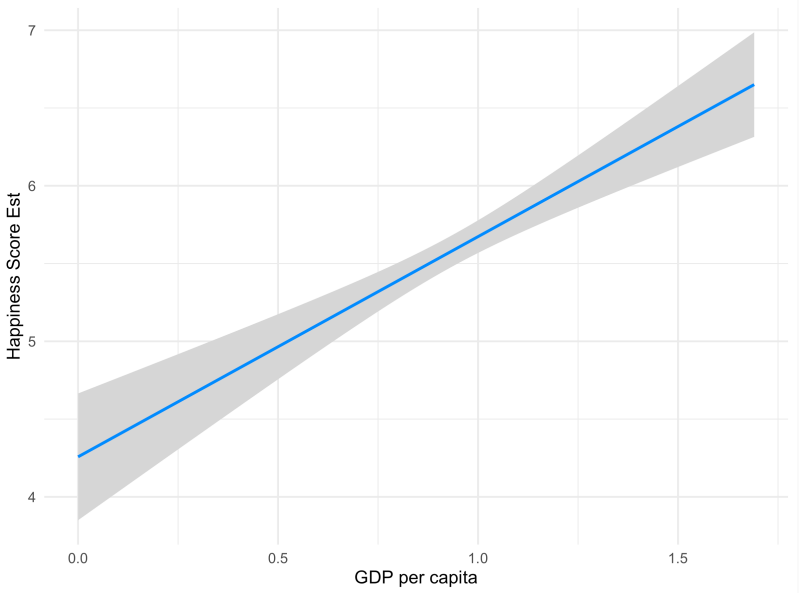
Visualize an interction: visreg()

Control other variables, visreg() will help you quickly view the regression relationship. `partial=FALSE` hides partial residuals. The band shows the confidence interval.

```
# install.packages("visreg")
library(visreg)

visreg(m2, "economy_gdp_per_capita",
               gg=TRUE, partial=FALSE,rug = FALSE) +
  labs(x="GDP per capita", y="Happiness Score Est")
```

Here gg=TRUE tells R to return a ggplot2 graph, so you can customize it using "+" and all other functions from ggplot2



Automated written model result summary: report()

Generates a written model summary. **Read and edit the wording, units, assumptions, and causal claims before using it in a report.** First, install package ---report

```
library(report)
report(m2)
```

```
We fitted a linear model (estimated using OLS) to predict happiness_score with
economy_gdp_per_capita, health_life_expectancy and freedom (formula:
happiness_score ~ economy_gdp_per_capita + health_life_expectancy + freedom).
The model explains a statistically significant and substantial proportion of
variance (R2 = 0.71, F(3, 154) = 128.20, p < .001, adj. R2 = 0.71). The model's
intercept, corresponding to economy_gdp_per_capita = 0, health_life_expectancy
= 0 and freedom = 0, is at 2.57 (95% CI [2.23, 2.90], t(154) = 15.14, p <
.001). Within this model:

  - The effect of economy gdp per capita is statistically significant and
positive (beta = 1.42, 95% CI [0.99, 1.84], t(154) = 6.61, p < .001; Std. beta
= 0.50, 95% CI [0.35, 0.65])
  - The effect of health life expectancy is statistically significant and
positive (beta = 0.95, 95% CI [0.27, 1.64], t(154) = 2.74, p = 0.007; Std. beta
= 0.21, 95% CI [0.06, 0.35])
  - The effect of freedom is statistically significant and positive (beta = 2.35,
95% CI [1.65, 3.05], t(154) = 6.63, p < .001; Std. beta = 0.31, 95% CI [0.22,
0.40])
```

Polynomial regression

For $y = \beta_0 + \beta_1 x + \beta_2 x^2$ the slope is $\beta_1 + 2\beta_2 x$

`I(x^2)` requests arithmetic squaring; `poly(x, 2)` uses orthogonal terms by default.

```
m_quad ← lm(happiness_score ~ economy_gdp_per_capita +
               I(economy_gdp_per_capita^2), data=WHR2016)
# Equivalent fitted curve; different coefficient parameterization:
m_poly ← lm(happiness_score ~
               poly(economy_gdp_per_capita, 2,raw=TRUE), data=WHR2016)
anova(m1, m_quad)
```

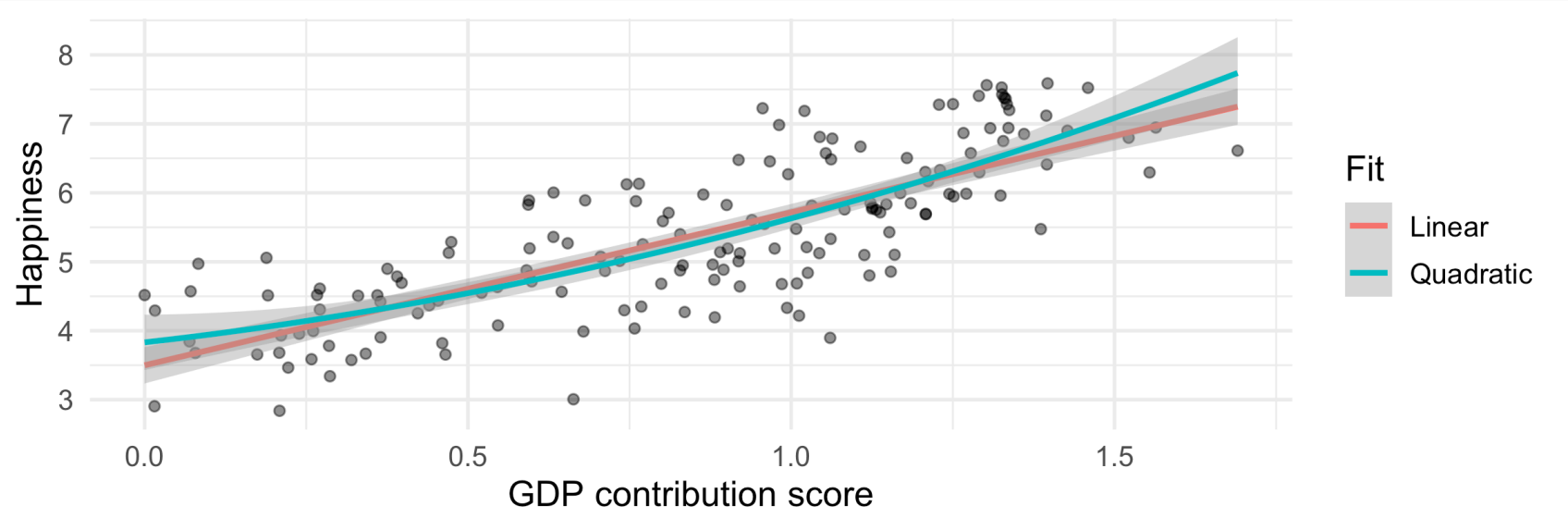
Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
156	80.3				
155	78	1	2.32	4.62	0.0331

Here anova() is used to compare two regression models and see whether adding the squared term significantly improves model fitness. The P-value is less than 0.05 and RSS(unexplained residuals) reduced, indicating that after adding the square term, the model has been improved.

Plot linear and quadratic fits

A curve can fit better without being causal. Avoid high-degree curves and extrapolation beyond the observed range.

```
ggplot(WHR2016, aes(economy_gdp_per_capita, happiness_score)) +
  geom_point(alpha=.5) +
  geom_smooth(aes(color="Linear"), method="lm",
              formula=y ~ x) +
  geom_smooth(aes(color="Quadratic"), method="lm",
              formula=y ~ x + I(x^2)) +
  labs(x="GDP contribution score", y="Happiness", color="Fit")
```



Regression with Fixed Effect - feols()

The `feols` command from package `fixest` allows for more flexibility in model specification, it estimates OLS with optional absorbed fixed effects and IV.

Fixed Effect:

> Fixed effects control for **characteristics shared within a group that might otherwise distort the estimated relationship.**

Commonly seen fixed effect for:

- Country fixed effect, which accounts for the stable differences between countries.
- Year fixed effect, which accounts for the common shocks affecting all countries in a given year.

Regression with Fixed Effect - feols()

feols(model equation, data, subset, vcov ...)

- **model equation:** an object of class "formula" containing a symbolic description of the model
- *subset:* an optional vector specifying a subset of observations to be used in the regression, for example, `subset = ~year ⩾ 2016`
- **data:** a data frame containing the variables indicated in the formula
- *vcov:* how standard errors are calculated, one of "iid", "hetero" (or "HC1"), "cluster", "twoway", or "conley", for example, `vcov = "cluster"`; `vcov= ~country`. etc.
- *cluster:* a list of vectors, a character vector of variable names, a formula or an integer vector specifying how to cluster standard errors
- ...

Regression with Fixed Effect - feols()

```
feols(y ~ x1 + x2, data=df, vcov="iid")

feols(y ~ x1 + x2 | region, data=df, vcov="hetero")

feols(y ~ x1 | country + year, data=panel, vcov=~country)
```

- `"iid"`: conventional standard errors.
- `"hetero"`: heteroskedasticity-robust standard errors.
- `~country`: country-clustered standard errors.

Regression with Fixed Effect - feols()

Try with the R dataset "census". Load package **tidyr**

> Estimate divorces as a function of population, urban population, and marriages, including region fixed effects and using standard errors clustered by state.

```
census ← census %>% select(state, region, divorce, pop, popurban, marriage) %>%
  drop_na()
# lm() can also add fixed effect by using factor()
reg1 ← lm(divorce ~ pop + popurban + marriage + factor(region), data=census)
reg2 ← feols(divorce ~ pop + popurban + marriage | region,
             data=census,
             vcov= cluster ~ state # this defines clustered std errors by state
            )
```

Region FE allow separate regional intercepts. The slopes use variation within regions.

Regression with Fixed Effect - feols()

```
census ← census %>% select(state, region, divorce, pop, popurban, marriage) %>%
  drop_na()
# lm() can also add fixed effect by using factor()
reg1 ← lm(divorce ~ pop + popurban + marriage + factor(region), data=census)
reg2 ← feols(divorce ~ pop + popurban + marriage | region,
             data=census,
             vcov= cluster ~ state # this defines clustered std errors by state
            )
summary(reg2)
```

```
OLS estimation, Dep. Var.: divorce
Observations: 50
Fixed-effects: region: 4
Standard-errors: Clustered (state)
         Estimate Std. Error  t value Pr(>|t|)
pop      0.000395   0.002120 0.186405 0.852897
popurban 0.003553   0.002278 1.559847 0.125232
marriage 0.183659   0.104001 1.765939 0.083636 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 6,257.6     Adj. R2: 0.927695
                Within R2: 0.935434
```

Pooled OLS and two-way fixed effects

Country FE absorb stable country differences; year FE absorb common year shifts.

```
panel ← whr_panel %>%
  select(country, year, happiness_score, economy_gdp_per_capita,
         health_life_expectancy, freedom) %>% drop_na()
m_pool ← feols(happiness_score ~ economy_gdp_per_capita +
                  health_life_expectancy + freedom,
                data=panel, vcov=~country)
m_fe ← feols(happiness_score ~ economy_gdp_per_capita +
                health_life_expectancy + freedom | country + year,
              data=panel, vcov=~country)
```

Add instrumental variables to feols()

Instrumental variables (IV) is included when an explanatory variable is correlated with unobserved factors (error term) affecting the outcome, which would making OLS potentially biased.

An instrument must satisfy:

- **Relevance:** it changes the endogenous explanatory variable.
- **Exogeneity:** it is unrelated to unobserved outcome determinants, conditional on controls.
- **Exclusion:** it affects the outcome only through that explanatory variable.

```
feols(y ~ x1 + x2 | fe1 + fe2 | x3 ~ iv3, data=df)
```

Here `x3 ~ iv3` uses instrument `iv3` for variable `x3`

The IV relationship always comes last after the fixed effect.

Add instrumental variables to feols()

Instrumental variables: two-stage least squares

Example: considering `popurban` as an instrument for `marriage` We want to estimate

$$Divorce_i = \alpha_{r(i)} + \beta_1 Pop_i + \beta_2 Marriage_i + \varepsilon_i$$

Here, i denotes a state, and $\alpha_{r(i)}$ represents its region fixed effect, and If $Marriage_i$ is correlated with the error term, OLS may be biased. Therefore, we adopt $Popurban_i$ as an instrument through **two-stage least squares (2SLS)**.

First stage: predict marriages

$$Marriage_i = \gamma_{r(i)} + \pi_1 Pop_i + \pi_2 Popurban_i + v_i$$

Obtain the predicted values, $\widehat{Marriage}_i$.

Add instrumental variables to feols()

Second stage: use predicted marriages

$$Divorce_i = \alpha_{r(i)} + \beta_1 Pop_i + \beta_2 \widehat{Marriage}_i + u_i$$

Key point: `popurban` enters the first stage but does not enter the second-stage outcome equation directly. The exclusion restriction requires urban population to affect divorces only through marriages, conditional on population and region fixed effects.

```
reg_iv ← feols(
  divorce ~ pop | region | marriage ~ popurban,
  data = census,
  vcov = "hetero")

summary(reg_iv, stage=1)
summary(reg_iv)
```

More on Result Output

Exporting regression tables

Choose the output around the task:

- `summary()`: inspect one fitted model.
- `modelsummary()`: compare models and format a report table.
- `huxreg()`: build and format a huxtable.
- `stargazer()`: text, HTML, or LaTeX for supported models.
- `etable()`: a convenient native table for `fixest` models.
- `broom::tidy(m)`: Extract coefficients | `broom::glance(m)`: Extract model statistics

Use huxreg()

install package `huxtable`

```
huxreg(reg1, reg2)
```

Argument	Purpose
Named models	Column headings
`coefs`	Select and relabel coefficients
`statistics`	Select model statistics
`number_format`	Decimal places
`stars`	Significance thresholds
`error_format`	SEs, t-statistics, or p-values
`note`	Explain uncertainty and model specification

Compare two model results

```
census_table ← huxreg(
  "OLS"=reg1, "Region FE"=reg2,
  statistics=c("N"="nobs", "R-squared"="r.squared"),
  number_format=3)
census_table
```

	OLS	Region FE
(Intercept)	-6381.693 *	
	(2650.694)	
pop	0.000	0.000
	(0.002)	(0.002)
popurban	0.004	0.004
	(0.002)	(0.002)
marriage	0.184 **	0.184
	(0.058)	(0.104)
factor(region)N Cntrl	7773.176 *	
	(3115.397)	
factor(region)South	11201.711 **	
	(3190.424)	
factor(region)West	8611.900 **	
	(3157.147)	
N	50	50
R-squared	0.937	0.937
*** p < 0.001; ** p < 0.01; * p < 0.05.		

Add labels and model notes

The models use different SE assumptions here: OLS conventional; FE heteroskedasticity-robust. State this in table notes.

```
census_table ← huxreg(
  "OLS"=reg1, "Region FE"=reg2,
  coefs=c(
    "Constant"="(Intercept)",
    "Population"="pop",
    "Urban population"="popurban",
    "Marriages"="marriage"),
    "North Central"="factor(region)N Cntrl",
    "South"="factor(region)South",
    "West"="factor(region)West"),
  statistics=c("N"="nobs", "R-squared"="r.squared"),
  number_format=3,
  stars=c("***"=.01, "**"=.05, "*"=.1),
  note=paste("Region FE: No / Yes. SEs: conventional /
             "{stars} Standard errors in parentheses.")
census_table
```

	OLS	Region FE
Constant	-6381.693 **	
	(2650.694)	
Population	0.000	0.000
	(0.002)	(0.002)
Urban population	0.004 *	0.004
	(0.002)	(0.002)
Marriages	0.184 ***	0.184 *
	(0.058)	(0.104)
North Central	7773.176 **	
	(3115.397)	
South	11201.711 ***	
	(3190.424)	
West	8611.900 ***	
	(3157.147)	
N	50	50
R-squared	0.937	0.937
Region FE: No / Yes. SEs: conventional / robust. *** p < 0.01; ** p < 0.05; * p < 0.1 Standard errors in parentheses.		

Use etable()

With etable(), results can also be displayed:

```
# Region entered as explicit dummy variables
reg1_feols ← feols(
  divorce ~ pop + popurban + marriage + factor(region),
  data = census
)

# Region entered through the fixed-effects section
reg2 ← feols(
  divorce ~ pop + popurban + marriage | region,
  data = census
)

etable(reg1_feols, reg2,
       dict=c("Constant"="(Intercept)",
            "Population"="pop",
            "Urban population"="popurban",
            "Marriages"="marriage",
            "North Central"="factor(region)N Cntrl",
            "South"="factor(region)South",
            "West"="factor(region)West")
       )
```

Table export exercise

> Create one comparison table, relabel predictors, identify the region FE, and export it.

```
quick_xlsx(census_table, file=here("census_model.xlsx"))

# Standalone LaTeX source; external LaTeX needed for PDF compilation:
quick_latex(census_table, file="output/census_models.tex")
```

`quick_xlsx()` needs the optional `openxlsx` package. Inspect the exported table for labels, uncertainty, model notes, and sample size.

More on modelsummary():labels, etc

We can also add confidence interval, P value, title, notes to the summary table, and change uncertainty estimation as well.

```
modelsummary(models, output="kableExtra",
  coef_map=coef_labels,
  vcov="HC1", conf_level=.95,
  fmt=3, stars=TRUE,
  statistic=c("SE = {std.error}", "p = {p.value}","[{co
  gof_map=c("nobs", "r.squared", "adj.r.squared"),
  title="Happiness associations, 2016",
  notes="95% intervals using HC1 robust standard errors
  escape = FALSE)
```

Happiness associations, 2016

	gdp/capita only	with controls
Constant	3.499***	2.569***
	SE = 0.120	SE = 0.180
	p = <0.001	p = <0.001
	[3.261, 3.736]	[2.213, 2.925]
GDP contribution score	2.218***	1.416***
	SE = 0.131	SE = 0.222
	p = <0.001	p = <0.001
	[1.959, 2.478]	[0.977, 1.855]
Health contribution score		0.953*
		SE = 0.382
		p = 0.014
		[0.198, 1.708]
Freedom contribution score		2.352***
		SE = 0.354
		p = <0.001
		[1.653, 3.050]
Num.Obs.	158	158
R2	0.610	0.714
R2 Adj.	0.607	0.708

95% intervals using HC1 robust standard errors

stargazer(): Console result, HTML, and LaTeX

Use supported `lm` objects here. `stargazer()` has no `type="kableExtra"`; use modelsummary or etable for `fixest` models.

```
library(stargazer)
stargazer(m1, m2, type="text",
  dep.var.labels="Happiness score",
  covariate.labels=c("GDP/capita", "Health", "Freedom"),
  omit.stat=c("f", "ser"))
# File outputs:
stargazer(m1, m2, type="html", out=here("models.html"))
stargazer(mtype="latex", out=here("models.tex"))
```

A publication table needs more than stars

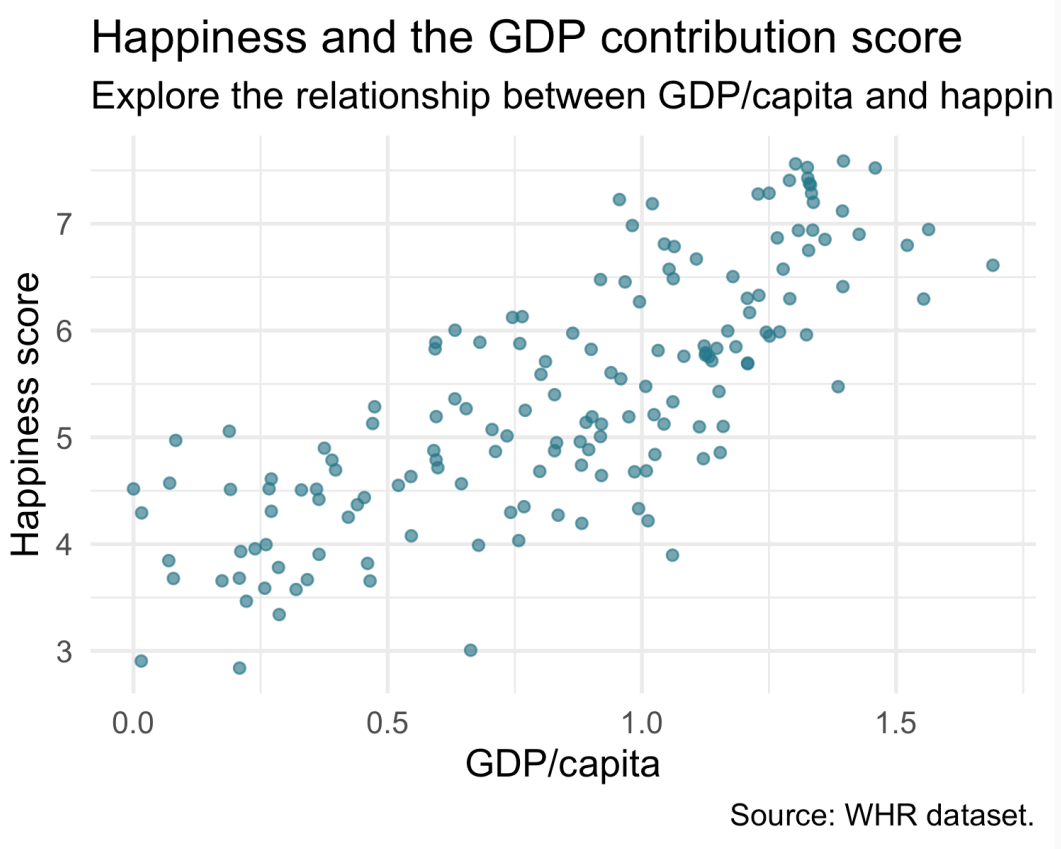
- Outcome name, units, and sample definition.
- Readable predictor labels and a stated reference category.
- Uncertainty: SEs or confidence intervals; identify robust/clustered choices.
- Number of observations and relevant fit measures.
- Fixed-effect indicators, data source, and model notes.
- Consistent samples when comparing specifications.

More on graph plotting

Continue from last session:labs() for ggplot()

`labs()` allows you amend titles, axis labels, legend titles, and notes for graph.

```
ggplot(WHR2016,
       aes(economy_gdp_per_capita, happiness_score)) +
       geom_point(color="#287C8E", alpha=.7) +
       labs(title=
              "Happiness and the GDP contribution score"
         subtitle=
         "Explore the relationship between GDP/capita ar
         x="GDP/capita", y="Happiness score",
         caption="Source: WHR dataset.") +
         theme_minimal(base_size=13)
```



`labs(color="Region")` **changes the title of legend**. Relabel category values with a scale or by recoding the variable.

ggsave(): export the plot you intended

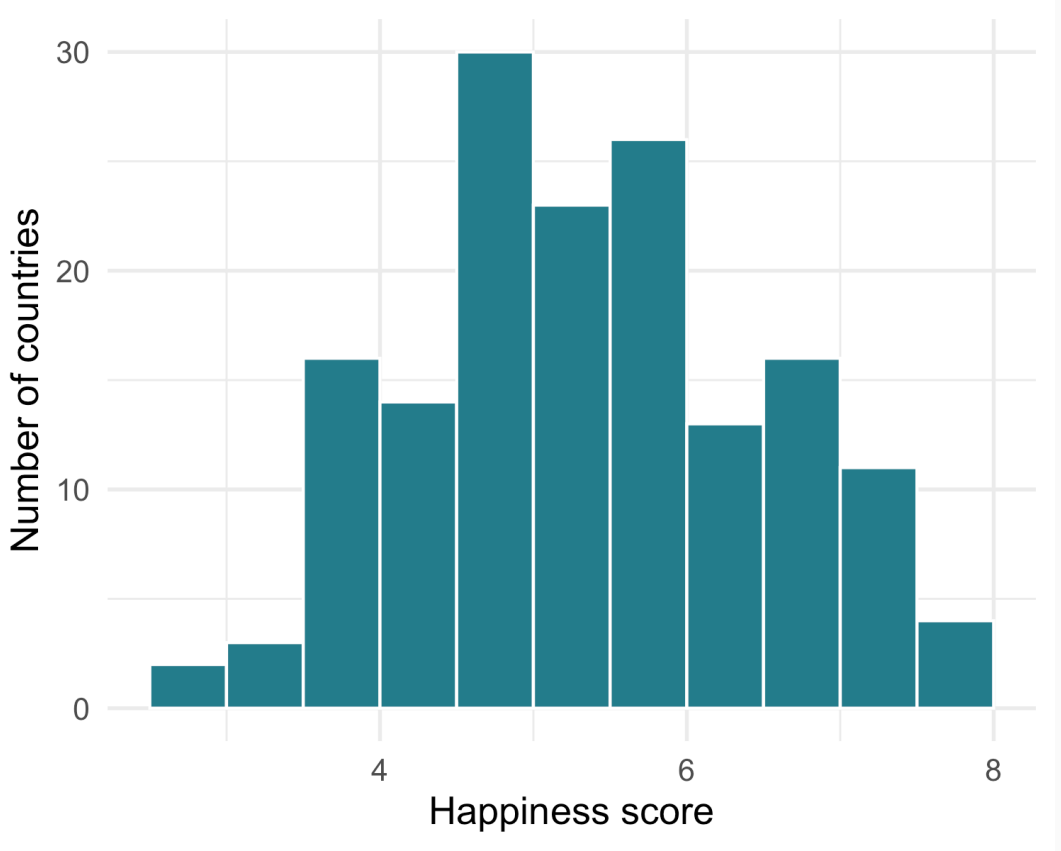
```
ggsave(filename=here("happiness_gdp.png"), plot=p,
       width=7, height=4.5, units="in", dpi=300, bg="white")
ggsave(filename=here(happiness_gdp.pdf"), plot=p,
       width=7, height=4.5, units="in")
```

- Express which plot you want to use by `plot=`; otherwise the last displayed plot is used.
- Choose physical dimensions for the report, then assess text readability.

Histograms: geom_histogram()

There is no standard ggplot2 `geom_histogram()`. Bin width changes the apparent distribution.

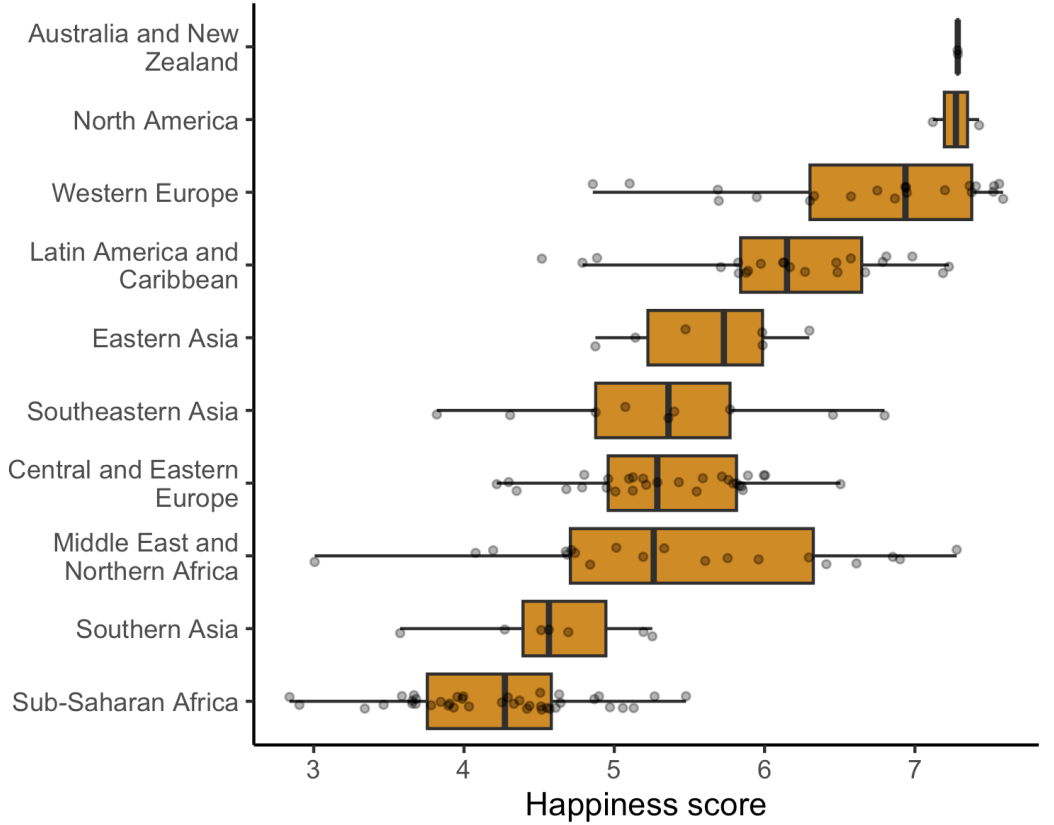
```
ggplot(WHR2016, aes(happiness_score)) +
  geom_hist(binwidth=.5, boundary=0,
                fill="#287C8E", color="white") +
  labs(x="Happiness score", y="Number of countries")
```



Box plots: compare distributions

Box: 25th–75th percentiles; line: median. Whiskers extend to observations within 1.5 IQR of the hinges; points beyond are outliers.

```
ggplot(WHR2016,
       aes(
         x=reorder(region, happiness_score, median),
         y=happiness_score)) +
  geom_boxplot(fill="#D38C28", outlier.shape=NA) +
  geom_jitter(width=.12, height=0, alpha=.35, size=1) +
  coord_flip() +
  labs(x=NULL, y="Happiness score")+
  theme_classic()
```

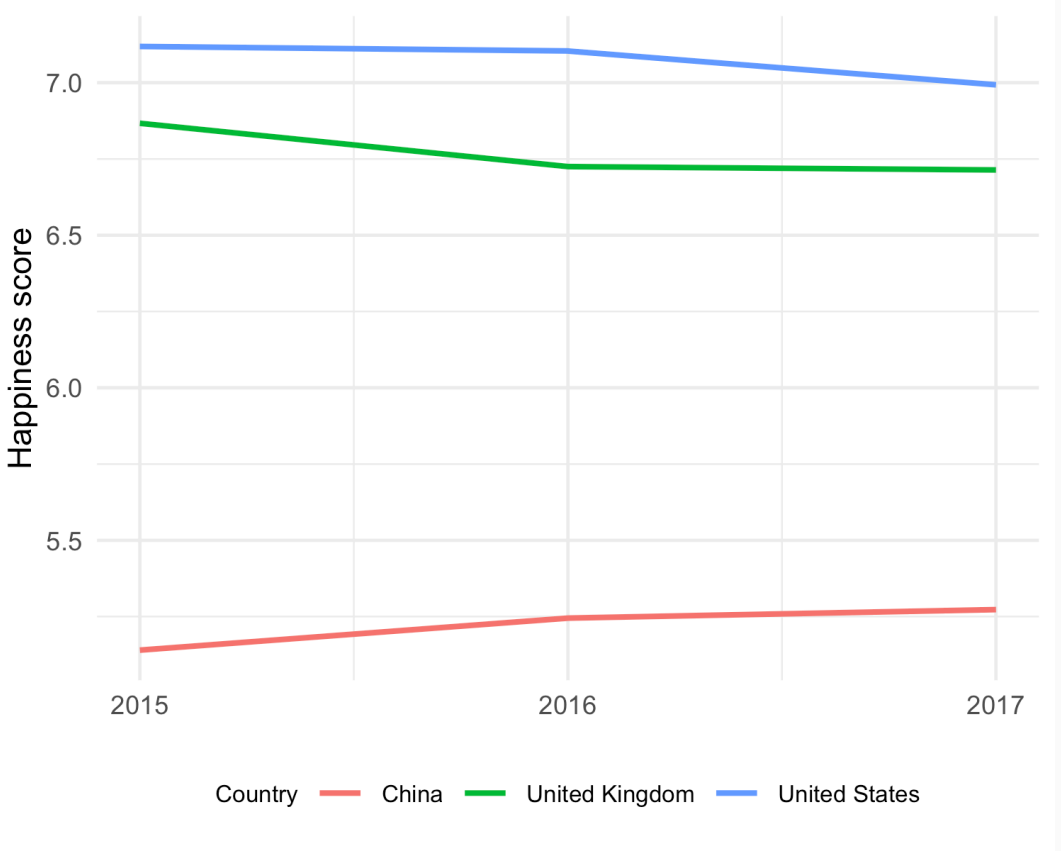


Line chart

Use `group=country` to connect the right observations. Line chart is often used for many years observation.

```
line_data ← whr_panel %>%
  filter(country %in%
         c("China", "United States", "United Kingdom"))
  %>%
  arrange(country, year)

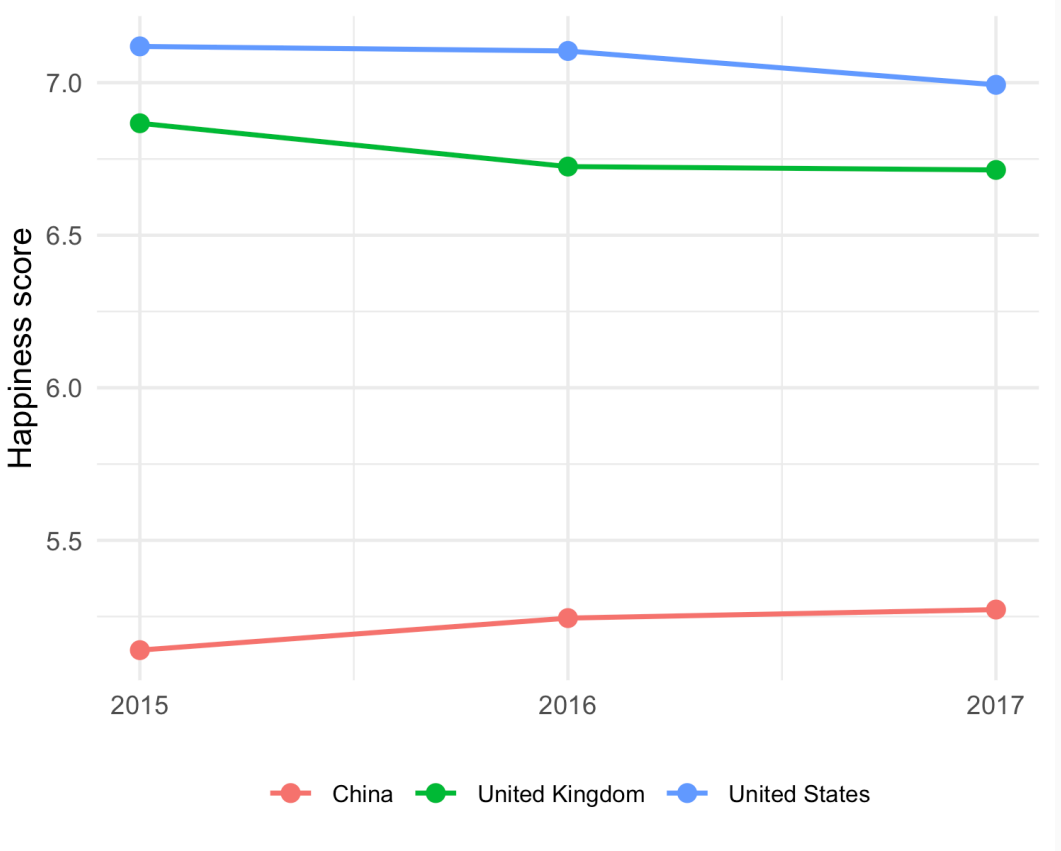
ggplot(line_data, aes(x=year,
                      y=happiness_score,
                      color=country,
                      group=country)) +
  geom_line(linewidth=.9) +
  scale_x_continuous(breaks=2015:2017) +
  labs(x=year, y="Happiness score",
       color="Country")
```



Line-dot chart: show observed dates

Dots identify observed annual values. A connecting line does not establish a continuous path between observations.

```
ggplot(chosen, aes(x=year,
                   y=happiness_score,
                   color=country,
                  group=country)) +
  geom_line(linewidth=.8) +
  geom_point(size=2.5) +
  scale_x_continuous(breaks=2015:2017) +
  labs(x=NULL, y="Happiness score", color=NULL) +
  theme_minimal() +
  theme(legend.position="bottom"
        legend.text=element_text(size=8))
```

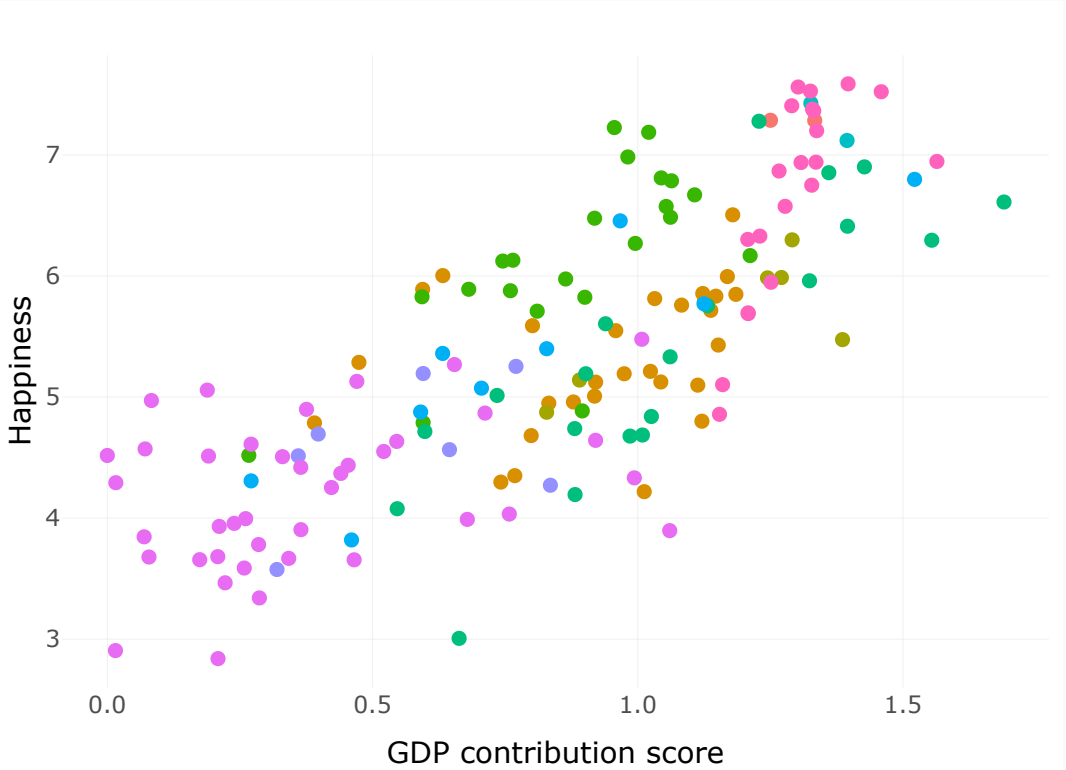


Try the interactive scatterplot

Package need: plotly()

```
interactive_p ← ggplot(WHR2016,
  aes(economy_gdp_per_capita, happiness_score,
      text=paste("Country:", country))) +
  geom_point(aes(color=region)) +
  labs(x="GDP contribution score", y="Happiness")
  ggplotly(interactive_p,
                tooltip=c("text", "x", "y"),
                width=NULL, height=384) %>%
    layout(showlegend = FALSE)
```

Can hover for country names, zoom into a range, or select legend entries.



Advanced: a 3-D scatterplot

Rotate to inspect three dimensions.

```
plot_ly(WHR2016,
  x=~economy_gdp_per_capita, y=~health_life_expectancy,
  z=~happiness_score, text=~country,
  type="scatter3d", mode="markers", width=NULL, height=
  marker=list(size=3, opacity=.7)) %>%
  layout(scene=list(xaxis=list(title="GDP score"),
    yaxis=list(title="Health score"),
    zaxis=list(title="Happiness")))
```

Appendix

Appendix: Command checklist

```
lm(y ~ ., data=selected_data)
lm(y ~ x1 * x2, data=df)
lm(y ~ x1:x2, data=df)
lm(y ~ x + I(x^2), data=df)
visreg::visreg(model, "x", gg=TRUE)
report::report(model)
fixest::feols(y ~ x | country + year, data=df, vcov=~country)
fixest::feols(y ~ control | endogenous ~ instrument, data=df)
modelsummary::modelsummary(list(m1, m2), output="kableExtra")
huxtable::huxreg(m1, m2)
```

Appendix: diagnostics and multicollinearity

```r
# Inspect residual patterns and influential observations:
par(mfrow=c(2, 2))
plot(m2)
par(mfrow=c(1, 1))

# Optional package, car:
car::vif(m2)
```

Highly overlapping predictors make individual coefficients less precise. A large VIF is a diagnostic signal, not an automatic deletion rule.

Residual diagnostics do not test away omitted-variable bias.

Appendix: exporting an interactive plot with package

```
library(htmlwidgets)
widget ← ggplotly(interactive_p)

saveWidget(widget,
  file=(here("happiness_interactive.html"), selfcontained=TRUE)
```

Self-contained widget export requires Pandoc and can create large files.

References and function help

- fixest: feols and IV
- huxtable: huxreg
- modelsummary documentation
- visreg documentation
- report documentation
- ggplot2 function reference
- Plotly R documentation

Takeaways

Takeaways

- Main effects, interactions, and polynomial terms answer different questions.
- Fixed effects change the comparison yet causal interpretation still needs assumptions.
- IV requires a justified instrument, not just a first-stage relationship.
- Report units, sample, uncertainty, and fixed effects clearly.
- Match the graph to the question and inspect the final exported file.